DEMONSTRATED CAPABILITY REPORT

Rahul Mitra

BlendED

Issued: November 2026

PROGRAM

Computer Vision and Robotics - Amazon Robotics Project

TOPIC

Image Enhancement

PROJECT LEAD

Dan Shao, Researcher in Computer Vision and Robotics

FORMAT & DURATION

Project-Based Learning (PBL) ·
8 weeks · 48 total research hours

This project builds computer vision systems for robotics, with students working in teams on a focused technical track and testing their methods on real image and video data. This topic focuses on recovering detail from degraded images. This team reproduced a U-Net-based super-resolution model, ran loss-function ablations, and tested robustness to varied blur across nature, text, and face images. Project Lead Dan Shao is VP of Computer Vision, Machine Learning and AI at ZELIG.

Demonstrated Capability

80 of 100 points

LEVEL 1
Foundational Application

LEVEL 2
Applied Reasoning

LEVEL 3
Independent Problem Solving

LEVEL 4
System-Level Insight **80** out of 80

SYSTEM-LEVEL INSIGHT · TIER DEFINITION

The learner operates with originality and self-direction, connects domain knowledge to broader system implications, and contributes ideas that advance the team beyond explicit prompts.

PROJECT LEAD JUSTIFICATION

Rahul combined substantial literature research with persistent, implementation-oriented follow-through and a strong willingness to test improvements. He was one of the biggest contributors to the project.

Participation & Attendance

20 of 100 points

Participation

10 / 10

Highly engaged and receptive to advice, making sustained efforts to try suggested approaches.

Attendance

10 / 10

No unexcused absences recorded. Present for all sessions.

Demonstrated Capability: 80 · Participation: 10 · Attendance: 10

FINAL SCORE **100 / 100**

The Demonstrated Capability tier framework is informed by the European Qualifications Framework for Lifelong Learning (European Commission). More information at europass.europa.eu.

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November 13, 2026

To Whom It May Concern:

My name is Dan Shao, and I am VP of Computer Vision, Machine Learning and AI at ZELIG. I acted as Rahul Mitra's Project Lead for Computer Vision and Robotics - Amazon Robotics Project, a Project Based Learning (PBL) program at BlendED. The project focuses on building vision systems for robotics, with student teams working in tracks that include stereo vision, image enhancement, and human posture detection.

Over the course of the eight-week online program, Rahul worked in the Image Enhancement track, where teams build and evaluate models that recover detail from degraded images. Rahul's team worked on single-image super-resolution, reconstructing high-resolution images from blurred, downsampled inputs. The team reproduced a U-Net-based super-resolution model, ran loss-function ablations, benchmarked results against published work using SSIM and PSNR across nature, text, and face images, and tested how well the model held up under varied blur. The team delivered a project website, a GitHub repository, and a demo video.

Rahul led much of the team's research into state-of-the-art super-resolution methods and the model approaches relevant to the project. He also shared the team's project website, GitHub repository, and demo video, the clearest record of the implementation the team delivered.

I worked with Rahul through our online sessions and team check-ins. Over that time, he demonstrated System-Level Insight, anticipating broader implications and downstream effects and connecting technical decisions to their real-world consequences. Rahul combined substantial literature research with persistent, implementation-focused follow-through. He explored SSIM and PSNR evaluation, loss ablations, blur conditions, and comparisons with a reference model. He was exceptionally hardworking and receptive to advice, making sustained efforts to try suggested approaches and improve the work, and he was one of the biggest contributors to the project. Rahul also kept strong attendance, participated actively in sessions, and worked well with his teammates.

Rahul's persistence and research depth carried much of the project. I recommend Rahul highly.

Should you have further questions about Rahul's participation in this project, you may contact BlendED at programs@blendedlearn.org for a quantitative breakdown of his performance in the project, as well as the full project itself.

Thank you for your time and consideration.

Sincerely,

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Dan Shao

Name: Dan Shao
Title: Researcher in Computer Vision and Robotics

Date: 2026-11-13

William Doolittle

Name: William Doolittle
Title: Academic Coordinator

Date: 2026-11-13

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